WORKED SOLUTION

# STEP II 2016, Question 4

Pure • Range of Rational Function

**STEP II 2016, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q4

## The key idea

Clearing the fraction turns the definition of $y$ into a quadratic equation in $x$, and "$x$ is real" becomes "the discriminant is non-negative". To get rid of $\theta$ we need an upper bound for $y\cos\theta - \sin\theta$ that holds for every $\theta$; writing it as a dot product of $(y,-1)$ with the unit vector $(\cos\theta, \sin\theta)$ gives exactly $\sqrt{y^2+1}$. In part (ii) every inequality in the chain must be an equality, and that pins down both $x$ and $\theta$.

## Solution

### Part (i)

First note that the denominator never vanishes: as a quadratic in $x$, $x^2 + x\cos\theta + 1$ has discriminant $\cos^2\theta - 4 < 0$, so it is always positive. Multiplying up is therefore safe:

$$y(x^2 + x\cos\theta + 1) = x^2 + x\sin\theta + 1 \iff (y-1)x^2 + (y\cos\theta - \sin\theta)\,x + (y-1) = 0.$$

If $y \neq 1$ this is a genuine quadratic in $x$, and it has a real root $x$ if and only if its discriminant is non-negative:

$$(y\cos\theta - \sin\theta)^2 - 4(y-1)^2 \geq 0.$$

If $y = 1$ the required inequality reads $(\cos\theta - \sin\theta)^2 \geq 0$, which is true anyway. So for real $x$,

$$(y\cos\theta - \sin\theta)^2 \geq 4(y-1)^2,$$

as required.

**Ask yourself** How can we remove $\theta$? We want something bigger than $(y\cos\theta - \sin\theta)^2$ that doesn't depend on $\theta$.

The expression $y\cos\theta - \sin\theta$ is the scalar product of $(y,-1)$ with the unit vector $(\cos\theta, \sin\theta)$, so its size is at most $|(y,-1)| = \sqrt{y^2+1}$. Equivalently, by the Cauchy–

Schwarz inequality, or directly from

$$(y^2+1)(\cos^2\theta+\sin^2\theta)-(y\cos\theta-\sin\theta)^2=(y\sin\theta+\cos\theta)^2\geq 0,$$

we have $(y\cos\theta-\sin\theta)^2\leq y^2+1$. Combining this with the first result,

$$y^2+1\geq(y\cos\theta-\sin\theta)^2\geq 4(y-1)^2,$$

which is the second inequality.

Expanding, $y^2+1\geq 4y^2-8y+4$, that is,

$$3y^2-8y+3\leq 0.$$

The roots of $3y^2-8y+3=0$ are $y=\dfrac{8\pm\sqrt{64-36}}{6}=\dfrac{4\pm\sqrt{7}}{3}$, and since the quadratic has positive leading coefficient it is non-positive exactly between its roots. Hence

$$\frac{4-\sqrt{7}}{3}\leq y\leq\frac{4+\sqrt{7}}{3}.$$

**Part (ii)**

Let $y=\frac{1}{3}(4+\sqrt{7})$, so $y-1=\frac{1}{3}(1+\sqrt{7})>0$. Then

$$y^2+1=\frac{16+8\sqrt{7}+7}{9}+1=\frac{32+8\sqrt{7}}{9},$$
$$4(y-1)^2=\frac{4(1+2\sqrt{7}+7)}{9}=\frac{32+8\sqrt{7}}{9}.$$

These are equal, and $2(y-1)>0$, so taking the positive square root gives $\sqrt{y^2+1}=2(y-1)$, as required. (Of course: $y$ is a root of $3y^2-8y+3=0$, which is the equation $y^2+1=4(y-1)^2$.)

**Ask yourself** What does this equality tell us about the chain of inequalities in part (i)?

The two ends of the chain $y^2+1\geq(y\cos\theta-\sin\theta)^2\geq 4(y-1)^2$ are now equal, so both inequalities are equalities.

*The discriminant is zero*, so the quadratic in $x$ has a repeated root

$$x=-\frac{y\cos\theta-\sin\theta}{2(y-1)}.$$

*The bound on $\theta$ is attained*, so $y\sin\theta+\cos\theta=0$, giving

$$\tan\theta=-\frac{1}{y}=-\frac{3}{4+\sqrt{7}}=-\frac{3(4-\sqrt{7})}{16-7}=\frac{\sqrt{7}-4}{3}.$$

(Here $\cos\theta \neq 0$, since otherwise $\sin\theta = 0$ too.) Moreover $(y\cos\theta - \sin\theta)^2 = y^2 + 1 = 4(y-1)^2$, so $y\cos\theta - \sin\theta = \pm 2(y-1)$ and therefore

$$x = \mp 1.$$

Concretely, $(\cos\theta, \sin\theta)$ is parallel to $(y, -1)$. If it points the same way, $\cos\theta > 0$ and $\sin\theta < 0$, then $y\cos\theta - \sin\theta = +2(y-1)$ and $x = -1$. If it points the opposite way, $\cos\theta < 0$ and $\sin\theta > 0$, then $x = 1$. In both cases

$$x = \pm 1, \qquad \tan\theta = \frac{\sqrt{7} - 4}{3}.$$

The two cases are really one: replacing $x$ by $-x$ and $\theta$ by $\theta + \pi$ leaves $y$ unchanged.

## Checks

- At $x = 1$ the formula is $y = \dfrac{2 + \sin\theta}{2 + \cos\theta}$. Taking $\theta$ in the second quadrant with $\tan\theta = \frac{1}{3}(\sqrt{7} - 4) \approx -0.4514$ gives $\theta \approx 2.7176$, and then $y \approx 2.21525$, which matches $\frac{1}{3}(4 + \sqrt{7}) \approx 2.21525$.
- Sampling many random real $x$ and $\theta$ by computer, the values of $y$ stayed between 0.45142 and 2.21524, just inside the bounds $\frac{1}{3}(4 \mp \sqrt{7}) \approx 0.451416$ and 2.215250.
- A neat coincidence to notice: $\tan\theta = -\frac{1}{3}(4 - \sqrt{7})$ is minus the lower bound, because the two bounds are reciprocals (their product is $\frac{1}{9}(16 - 7) = 1$).

## Common slips

- Forgetting the case $y = 1$, where the equation in $x$ is not a quadratic and the discriminant argument does not apply directly.
- Taking $\sqrt{y^2 + 1} = -2(y-1)$ or not checking signs: the positive root is correct here only because $y > 1$.
- Giving only one value of $x$; both $x = 1$ and $x = -1$ occur, with $\theta$ in different quadrants.

## Beyond STEP

The step $(y\cos\theta - \sin\theta)^2 \leq (y^2 + 1)(\cos^2\theta + \sin^2\theta)$ is the two-dimensional case of the Cauchy–Schwarz inequality $|\mathbf{u}\cdot\mathbf{v}| \leq |\mathbf{u}|\,|\mathbf{v}|$, with equality exactly when the vectors are parallel. This inequality holds in every inner-product space and is one of the most used tools in first-year analysis and linear algebra. Strang's *Introduction to Linear Algebra* develops it geometrically; Abbott's *Understanding Analysis* shows it at work in analysis.

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